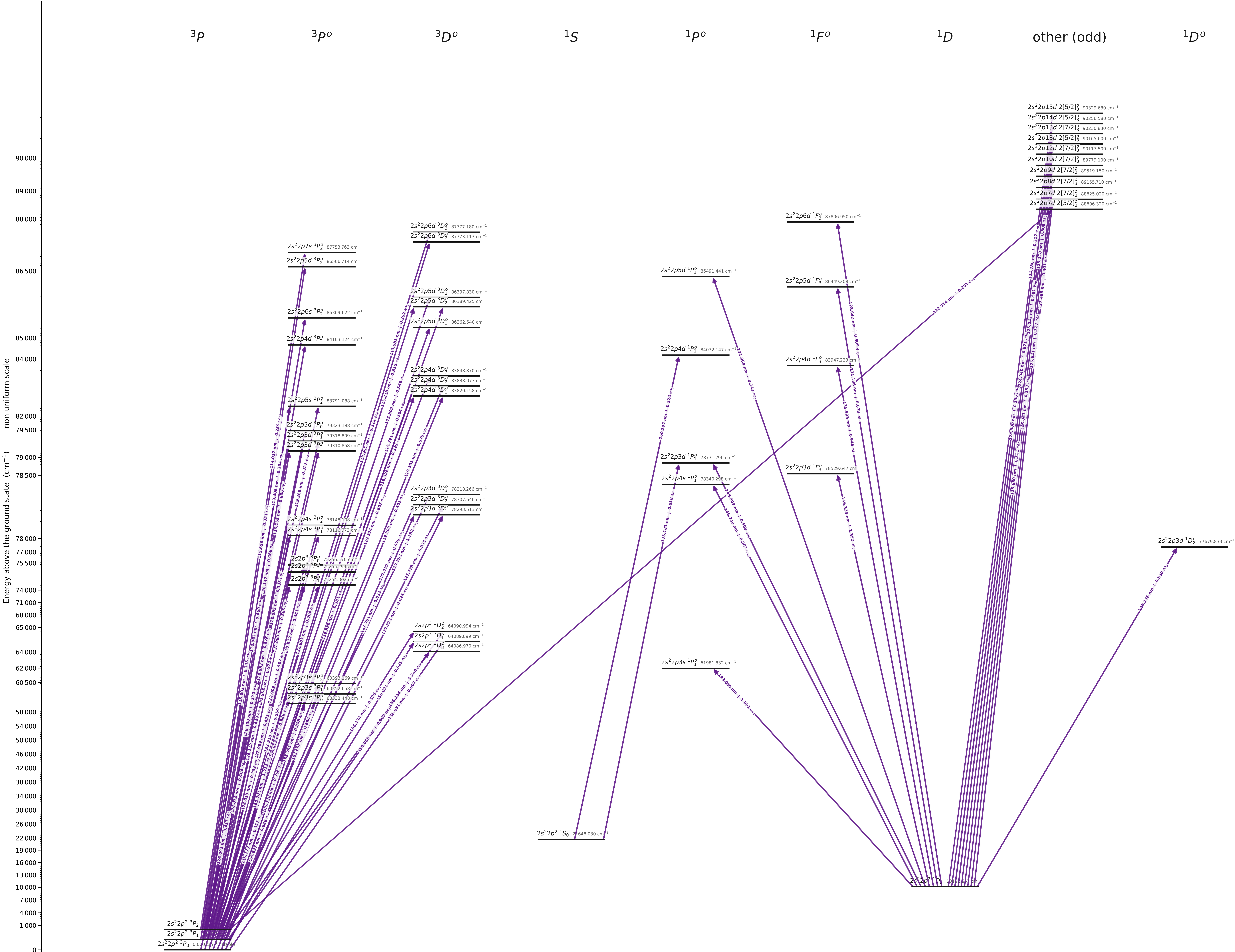
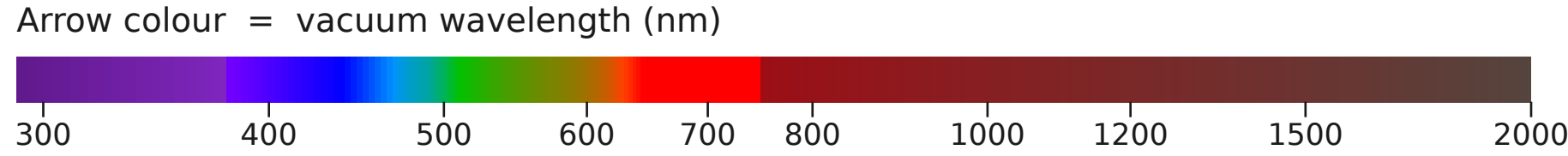


Carbon

C (C I). Strongest classified transitions below 2 μm (NIST): 55 levels, 70 lines.

C⁺ ionisation limit 90820.35 cm⁻¹ = 11.26029 eV (110.107 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
2s ² 2p ² ³ P ₂ – 2s ² 2p4d ³ D ₃ ^o	119.3240	–	2512.4244	0.807	1.110e+08	–	–	–
2s ² 2p ² ³ P ₁ – 2s ² 2p3d ³ P ₀ ^o	126.0926	–	2377.5572	0.417	1.760e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s ² 2p3d ³ P ₂ ^o	126.1552	–	2376.3785	0.806	1.310e+08	–	–	–
2s ² 2p ² ³ P ₀ – 2s ² 2p3d ³ D ₁ ^o	127.7245	–	2347.1805	0.624	1.260e+08	–	–	–
2s ² 2p ² ³ P ₁ – 2s ² 2p3d ³ D ₂ ^o	127.7282	–	2347.1120	0.935	1.700e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s ² 2p3d ³ D ₃ ^o	127.7550	–	2346.6210	1.282	2.280e+08	–	–	–
2s ² 2p ² ³ P ₁ – 2s2p ³ ³ P ₀ ^o	132.9085	–	2255.6310	0.507	2.220e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s2p ³ ³ P ₂ ^o	132.9577	–	2254.7955	0.975	1.640e+08	–	–	–
2s ² 2p ² ¹ D ₂ – 2s ² 2p4d ¹ F ₃ ^o	135.5848	–	2211.1063	0.946	1.040e+08	–	–	–
2s ² 2p ² ¹ D ₂ – 2s ² 2p3d ¹ F ₃ ^o	146.3336	–	2048.6914	1.392	1.790e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s2p ³ ³ D ₃ ^o	156.1437	–	1919.9775	1.240	1.170e+08	–	–	–
2s ² 2p ² ³ P ₀ – 2s ² 2p3s ³ P ₀ ^o	165.6928	–	1809.3272	0.884	1.160e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s ² 2p3s ³ P ₂ ^o	165.7008	–	1809.2402	1.712	2.610e+08	–	–	–
2s ² 2p ² ³ P ₁ – 2s ² 2p3s ³ P ₀ ^o	165.7907	–	1808.2591	0.883	3.470e+08	–	–	–
2s ² 2p ² ³ P ₂ – 2s ² 2p3s ³ P ₂ ^o	165.8121	–	1808.0257	0.986	1.440e+08	–	–	–
2s ² 2p ² ¹ D ₂ – 2s ² 2p3s ¹ P ₁ ^o	193.0905	–	1552.6004	1.901	3.390e+08	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).